

BRANDON REISS

Summit, NJ (NYC metro) | brandonreiss.com | LinkedIn

SUMMARY

Applied AI systems architect and former founder specializing in production ML platforms that handle real-world scale, latency, and cost constraints. Proven track record of operating autonomously to translate ambiguous R&D bets into durable production systems, architecting high-throughput unstructured data intelligence layers, and advising executive leadership on company-wide AI strategy.

EXPERIENCE

HubSpot | Remote

Jan 2025 – Present

Staff Software Engineer

- Architect and top contributor for HubSpot's unstructured data intelligence layer processing ~200B tokens per day and ~30M globally distributed record update triggers (>600 RPS peak) across 60M orchestrated extraction tasks, while maintaining p99 end-to-end latency under 4 minutes and unit costs below \$0.001 per record.
- Integrated unstructured context into flagship AI assistant, setting evaluation standards from experimentation to release; achieved >30% daily traffic penetration with measurable lifts in CSAT and model quality benchmarks.
- Designed company-wide unstructured content normalization and noise-filtering standards, significantly cutting inference token waste, reducing hallucinations, and establishing sustainable unit economics for LLM features.
- Aligned unstructured intelligence platform engineering with executive leadership's top strategic priority of leveraging AI context to deliver measurable customer outcomes by influencing roadmap priorities and developing new AI context outcome measurement methods.
- Influenced staff+ engineers to invest in agentic AI performance optimization using replay-driven evaluation frameworks; drove cross-team coaching initiatives, including accelerating stalled product launches and connecting high-impact knowledge transfer from Customer Success to adjacent MLE teams.

Frame AI | New York, NY

Feb 2016 – Jan 2025 (Acquired by HubSpot)

Co-Founder / Principal Engineer / Director

- Co-founded and served as principal architect from inception to acquisition by HubSpot, scaling an applied AI platform over 4 orders of magnitude to handle millions of monthly conversations powering dozens of process automations at a dozen mid-market and enterprise clients.
- Provided technical leadership to lean engineering organization of 6 across full-stack, data engineering, R&D, and forward-deployed functions, delivering major yearly product releases, weekly customizations, and a major white-label product integration with Twilio Flex.
- Architected multi-stage ingestion, extraction, and synthesis pipelines processing continuous streams of customer interactions under demanding end-to-end SLAs.
- Implemented task prioritization and deduplication strategies across heterogeneous computational workloads and multi-tenant fault isolation across service fleet.
- Synthesized classical NLP and modern ML into configurable, cost-efficient production pipelines—unifying rules-based filtering, dependency parsing, sentence encoders, and LLMs into cohesive intelligence systems.
- Engineered storage backends combining relational databases (PostgreSQL) for transactional state with columnar and distributed object/NoSQL storage (ClickHouse) for high-volume analytical workloads.
- As Head of Security, successfully unlocked Fortune 100 go-to-market strategy by establishing comprehensive security, compliance, and risk management programs; achieved and maintained SOC 2 Type II certification for 5 consecutive years with zero notable exceptions, delivered HIPAA compliance, and cleared security audits.

Knewton | New York, NY

Sep 2012 – Dec 2015

Software Engineering Manager / Senior Engineer

- Selected by senior leadership to lead a high-performing team of 8 engineers within the Adaptive Learning division, bridging core R&D with production engineering and setting technical execution standards.
- Built low-latency (p99 < 2 minutes) statistical inference engines and recommendation systems serving millions of students globally.

Organic Motion | New York, NY

Jan 2007 – Sep 2012

Software Engineer

- Developed low-latency computer vision algorithms and local network distributed systems for markerless motion capture and clinical gait analysis.
- Self-taught hardware and firmware development to produce custom PCBs mounted with rings of near-IR LEDs and programmable microcontroller.

TECHNICAL EXPERTISE & CORE COMPETENCIES

Agentic AI & Compound Systems → Token Budgeting & Context Window Management, Tool & Skill Architectures, Agent Task Performance Evaluation & Benchmarking, Task Aligned Data Retrieval, RAG.

Applied AI / ML & Statistical Inference → Recommender & Ranking Systems, Multi-Class Text Classification, Generalized Linear Models (GLMs), Signal Extraction & Feature Engineering, Sentence Encoders, Clustering, Syntactic/Dependency Parsing, Classical NLP & Rule-Based Heuristics.

Distributed Systems & Concurrency → Low-Latency Event-Driven Architectures, Multi-Tenant Resiliency (Blast-Radius Isolation, Anti-Correlated Failure Design), Priority-Tiered Processing, Fault-Tolerant Distributed Queues & Worker Architectures, Asynchronous / Parallel / Concurrent Execution Patterns, CPU & Memory Profiling.

Data Infrastructure & Storage → Relational Databases (PostgreSQL & InnoDB Internals, Concurrency, Lock Mechanics, Indexing), Columnar OLAP Datastores, Distributed Storage (HBase, S3).

Security, Compliance & Governance → SOC 2 Type II Program Architecture & Audit Defense, HIPAA Compliance, Enterprise Security Reviews, Risk Management & Data Governance, Application Security Testing.

Languages & Technical Tooling → Python, C++, Java, SQL, Bash / Linux CLI Tools; broad polyglot proficiency across modern software stacks; agentic coding harnesses & AI-assisted development workflows.

EARLY CAREER & APPLIED RESEARCH

Undergraduate Research Assistant

Rensselaer Polytechnic Institute (RPI)

Co-authored peer-reviewed research in Applied and Environmental Microbiology (April 2006, 72: 2460–2470) on isolating PCB-dechlorinating bacterial cultures using PCR, DNA sequence analysis, and gas chromatography.

Engineering Intern

BAE Systems

First high-school intern admitted in company history; designed a \$300K RF-absorbent antenna test facility and conducted experimental classification of RF signal polarization using neural networks.

EDUCATION

M.S. in Computer Science

New York University, Courant Institute of Mathematical Sciences

Recipient, Courant Hedwig Kurzbart Master's Fellowship (awarded annually for academic excellence)

B.A. in Computer Science and Mathematics

New York University, College of Arts and Science

Minor in Metropolitan Studies